ARUN DAS

AI / ML Engineer · Data Scientist · MCA 2026

itsmearundasofficial@gmail.com | +91-7736908820 | Kattappana, Kerala, India | Portfolio | GitHub

PROFESSIONAL SUMMARY

MCA (Data Science & AI) graduate who ships - not just studies. Built and deployed a hybrid YOLOv8 + EfficientNet-B0 detection pipeline with a live demo and built an end-to-end credit default prediction system comparing Random Forest and XGBoost. Skilled in Python, PyTorch, scikit-learn, and OpenCV, with Flask/React deployment experience and hands-on training in AWS, Docker, and CI/CD. Seeking an entry-level AI/ML Engineer / Data Scientist role to bring the same bias toward shipped, working models.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+)

ML / AI: YOLOv8, EfficientNet-B0, ByteTrack, PyTorch, Random Forest, XGBoost, scikit-learn, SMOTE, OpenCV

Backend & APIs: Flask, RESTful APIs, Node.js, Express.js

Frontend: React.js, HTML5, CSS3

Databases: MongoDB, MySQL, SQLite

Cloud & Tools: AWS (EC2, S3, IAM), Docker, Git/GitHub, CI/CD Pipelines, VS Code

PROJECTS

Hybrid Vision AI - Object Detection & Classification System *Jan 2026 – Mar 2026*

Live Demo: itsmearundas.github.io/Detecting-and-classifying-multiple-objects-in-images-and-videos/

GitHub: github.com/itsmearundas/Detecting-and-classifying-multiple-objects-in-images-and-videos

Python · PyTorch · YOLOv8n · EfficientNet-B0 · ByteTrack · Flask · React.js · OpenCV

- Engineered a two-stage deep-learning pipeline - YOLOv8n for real-time localization, fine-tuned EfficientNet-B0 for classification - reaching **85.41% test accuracy and 0.85 macro-F1** across 17 CIFAR-100 classes.
- Integrated ByteTrack multi-object tracking behind a Flask REST API benchmarked at **43 fps (GPU) / 18 fps (CPU)**, with live bounding-box and confidence overlays.
- Built a responsive React frontend with image/video/webcam mode-switching and zero-refresh, real-time inference rendering via continuous frame streaming.

Credit Card Default Prediction System *Jun 2025 – Dec 2025*

GitHub: github.com/itsmearundas/Credit-Card-Default-prediction

Python · Random Forest · XGBoost · Scikit-learn · SMOTE · Flask · React.js · Node.js · MongoDB

- Built an end-to-end credit default prediction platform, comparing Random Forest and XGBoost on 30,000 records from the UCI Taiwan Credit Default dataset.
- Handled class imbalance with SMOTE, tuned models with GridSearchCV, and set up a SHAP-ready pipeline for model interpretability.
- Integrated the trained model into a web application using Flask, React.js, Node.js, and MongoDB.

ExpenseTracker - Full-Stack Personal Finance Manager *2026*

Live Demo: expensetracker-33dk.onrender.com/ | GitHub: github.com/itsmearundas/ExpenseTracker

Python · Flask · SQLite · JavaScript · HTML5 · CSS3

- Built a full-stack personal finance manager with a Flask + SQLite backend and a vanilla JavaScript frontend, supporting real-time create, read, update, and delete of expense records.
- Added category-based spend analytics to summarize spending patterns, and deployed the app live on Render.

EDUCATION

Master of Computer Applications (MCA) - Data Science & AI *Aug 2024 – May 2026*

Mar Athanasius College of Engineering · APJ Abdul Kalam Technological University, Kerala CGPA: 7.99 / 10

CERTIFICATIONS & TRAINING

- Cloud Computing Workshop (5 days) - IIIT Kottayam × Educ Kshetra, Aug 2025 - AWS EC2, S3, IAM, Docker, CI/CD pipelines, VM management.